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Tract-specific magnetization transfer ratio provides insights into the severity of degenerative cervical myelopathy

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Abstract

STUDY DESIGN: Cross-sectional study.

OBJECTIVES: This study's goal is to report whether Magnetization Transfer Ratio (MTR) can evaluate the severity of white matter (WM) injury in degenerative cervical myelopathy (DCM).

SETTING: Laureate Institute of Brain Research, USA; Department of Neurosurgery, University of Oklahoma Health Sciences Center, USA.

METHODS: 27 DCM patients were aged-matched with 20 healthy controls (HC) and categorized into treatment groups based on modified Japanese Orthopedic Association (mJOA) severity (11 mild and 16 moderate/severe). Regional and tract MTRs were extracted from the two vertebral levels containing maximum compression within magnetization transfer images. MTR differences

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AUTHOR CONTRIBUTIONS

Concept/Idea/Design: GH, FM, TP, YD, KAW, ZAS; Data Collection: Data Collection: GH, FM, AFK, SH; Data Analysis: FM, GH; Writing: GH, FM, LD, KAW, ZAS; Resources: HS, MVH, DD, MR, ZAS.

CODE AVAILABILITY

Due to the ongoing nature of the analysis of the patient datasets, the code for this analysis is not yet publicly available. However, until such time that the code used for this analysis is made publicly available on GitHub, it is available upon reasonable request.

COMPETING INTERESTS

The authors declare no competing interests.

between groups were assessed using a one-way ANOVA or Kruskal-Wallis test. The association between MTR and mJOA measures was evaluated using Spearman's correlation.

RESULTS: Significant decreases in MTR were found between HC and moderate/severe groups in the overall ($p = 0.0065$) and ventral ($p = 0.0009$) WM regions; and ventral corticospinal ($p = 0.0101$), ventral reticulospinal ($p = 0.0084$), spinal lemniscus ($p = 0.0079$), and fasciculus cuneatus ($p = 0.0219$) tracts. The spinal lemniscus MTR also significantly decreased between HC and mild groups ($p = 0.038$). Ventral reticulospinal tract MTR correlated with upper ($r = 0.439$; $p = 0.022$) and lower ($r = 0.386$; $p = 0.047$) limb motor mJOA scores.

CONCLUSIONS: Significant tract-based MTR changes and correlations align with known DCM symptoms, are demonstrated to be lost at the regional level, and display the inhomogeneous compressive damage occurring within DCM spinal cords.

INTRODUCTION

Degenerative cervical myelopathy (DCM) is an age-related, degenerative spinal cord condition impacting the cervical spine that, if left untreated, can result in chronic injury to the spinal cord. Common symptoms of DCM can include numbness and loss of motor coordination in the hands and fingers, trouble balancing, and chronic neck pain [1, 2], which begins to worsen and impede quality of life as the myelopathy progresses. The most commonly used clinical grading assessment scale for DCM severity is the modified Japanese Orthopedic Association (mJOA) survey [1, 2]. The mJOA survey reflects common DCM symptoms that manifest as a result of compressive damage in the spinal cord and has been demonstrated to reliably classify mild, moderate, and severe myelopathy by assessing upper and lower limb motor dysfunction, upper limb sensory loss, and sphincter dysfunction [3]. Though multiple treatments and therapies exist [1, 2], DCM compression can only be permanently corrected with surgical decompression [4, 5], typically recommended for moderate and severe cases, but it cannot guarantee neurological recovery [6–8]. Symptom duration has been shown to be an important factor in predicting post-surgical recovery because it is theorized to reflect permanent damages resulting from demyelination of spinal cord white matter [6, 9]. The exclusion of mild DCM patients as surgical candidates potentially leaves patients in the early stages of DCM vulnerable to worsening symptoms [8], however whether there are white matter differences between non-surgical (i.e., mild DCM) and surgical (i.e., moderate and severe DCM) groups is unknown.

There are several imaging methodologies that can be used to indirectly measure myelin levels within the spinal cord, such as diffusion tensor imaging [10, 11], myelin water fraction [12], and synthetic MRI [13]. One of these indirect methodologies is Magnetization Transfer Ratio (MTR), which can be calculated from structural MRI and has already been applied to study white matter damage in spinal cord pathologies like multiple sclerosis [14] and spinal cord injury [15]. MTR measures within DCM patients have been shown to be lower than healthy controls [16, 17], which reflects the white matter tract volume loss associated with the manifestation of DCM symptoms [18]. Early MTR studies have also shown that ventral spinal cord MTR is associated with pre-operative mJOA/JOA scores [17, 19] and post-operative mJOA recovery [20]. Additionally, MTR from the rubrospinal tract was found to be associated with post-operative mJOA score recovery [20], as were

spinocerebellar MTR tract values and pre-operative JOA scores [19]. However, these studies pose several design gaps that prevent them from impacting clinical intervention decision making. Firstly, the exclusion of key participant groups in some of these studies, e.g., clinically severe myelopathy patients [20] and healthy controls [19], limits the scope of the results because they can only be applied to specific clinical treatment groups. Secondly, only a limited amount of studies performed tract-based MTR extraction [19, 20], which was further technically limited by their primary reliance on hand drawing spinal cord masks [17, 20] instead of automated cord segmentation. Therefore, the design and technical gaps within these studies prevent a full investigation into MTR's ability to reflect white matter damage in DCM severity at the individual tract level.

In the present study, we build upon these MTR studies to further investigate whether MTR has the ability to quantify the different severity levels of white matter damage in clinical DCM patient groups by using an automated, deep learning based spinal cord segmentation tool to focus on tract-based white matter analysis. To do so, we compared regional- and tract-based MTR between mJOA-based DCM treatment groups and their correlation to the clinical mJOA measures. Because white matter damage occurs as myelopathy continues to progress, MTR is expected to significantly differ between HC and DCM severity groups. Furthermore, because white matter damage leads to the impairment of the spinal cord's regions and tracts, regional- and tract-based MTRs are expected to correlate to mJOA measures relating to their corresponding sensory and motor functions.

METHODS

Participants

Convenience sampling was used to recruit degenerative cervical myelopathy patients (n = 27, 15 female and 12 male) from the OU Health Physicians Neurosurgery Clinic. All DCM patients received a DCM diagnosis from a licensed clinician and had an mJOA score less than 18 (Table 1). Aged matched healthy control participants (n = 20, 14 female and 6 male) were recruited from the Laureate Institute for Brain Research (LIBR) in Tulsa, OK and the University of Oklahoma Health and Science Center (OUHSC). All participants were scanned between 2021 and 2023 and a signed, informed consent was collected before any data collection.

The following inclusion and exclusion criteria for this study is further described by Muhammad et al. [21]. Inclusion criteria for DCM patients included evidence of cervical spinal cord compression on a T2-weighted image and one of more of the following: (1) hyperreflexia, (2) a positive Hoffman sign, (3) hand coordination loss, (4) gait impairment, and/or (5) motor weakness. Furthermore, the following exclusion criteria were applied to all participants: (1) participant is less than 30 or more than 70 years of age, (2) a history of spine surgery, (3) three or more co-morbid conditions, (4) diagnosis of a neurodegenerative disease, (5) diagnosis of a systemic rheumatological disease, (6) presences of isolated radiculopathy without myelopathy, (7) a history of active peripheral/vascular neuropathy, (8) pregnancy, (9) a history of seizure disorders, (10) any medical condition preventing the completion of study related procedures, (11) DCM patients in need of urgent decompressive surgery, and/or (12) participants with asymptomatic spinal cord compression (characterized

by the presence of cervical canal stenosis, disk herniation, or hyperintense signal on a T2-weighted image, with no manifestation of clinical symptoms and an mJOA score of 18).

Statement of ethics

This study was approved by the University of Oklahoma Health Sciences Center institutional review board (IRB #12068) and is in accordance with the Declaration of Helsinki. We certify that all applicable institutional and governmental regulations concerning the ethical use of human volunteers were followed during the course of this research.

Clinical assessment scales

All participants filled out a modified Japanese Orthopedic Association (mJOA) survey [3]. An mJOA score is broken into four sections, measuring four independent factors: upper extremity motor dysfunction (0–5), lower extremity motor dysfunction (0–7), sensation (0–3), and sphincter dysfunction (0–3). Answers for each question were summed and each participant was given a score of 0–18. DCM participants were then labeled as having mild, moderate, or severe myelopathy based on the mJOA severity groups established by Tetreault et al. [3]. Mild DCM was defined as a score of 15–17, moderate DCM as score of 12–14, and severe DCM as a score of 0–14 [3]. DCM participants were sorted into mild ($n = 11$, 8 female and 3 male) and moderate/severe ($n = 16$, 7 female and 9 male) groups based on their resulting mJOA scores to simulate treatment groups (i.e. non-surgical versus surgical). Healthy controls all received scores of 18, indicating the absence of neurological dysfunction. However, healthy controls with an mJOA of 18 were excluded if radiological signs of asymptomatic compression were present.

MRI protocol

All MRI data was collected at LIBR (Tulsa, OK) with a 3 T MR750 GE scanner equipped with a 16-channel NovaCSpine coil. Participants were scanned in a supine position within the scanner and instructed to remain as still as possible. Spinal cord image acquisition for healthy controls and DCM patients followed the generic spine consensus protocol [22]. T2-weighted scans (TR = 120 ms; TE = 2500 ms; FOV = 256 mm x 256 mm; Thickness = 0.8 mm) were collected for all participants. Additionally, two gradient echo T2*-weighted scans (TR = 35 ms; TE = 4 ms; Flip Angle = 9°; FOV = 172 mm x 172 mm; Resolution = 230 mm x 230 mm; Thickness = 5 mm) were collected with (MT1) and without (MT0) a magnetization transfer (MT) pulse. The MT scans are centered at the C5 vertebra and cover the spinal cord region from the C3 to C7 vertebral levels.

MRI analysis

Quantitative MT image analysis from each participant was performed using the Spinal Cord Toolbox (SCT v6.1) [23]. Automatic segmentation of the spinal cord was performed on the MT1 images utilizing SCT's deep learning model (*sct_deepseg_sc* [24]). The accuracy of these segmentation masks was verified through a quality control report and manual corrections were applied to optimize automatic segmentation outcomes. A binary spinal cord mask with a diameter of 35 mm was created on the MT1 image to define a region of interest (ROI) surrounding the spinal cord to avoid interference from irrelevant body

structures during the registration of the MT1 to MT0. The MT0 image was then registered to the MT1 using *sct_register_multimodal* and the ROI mask with a regularized slice-wise translation [25]. MTR was calculated using the *sct_compute_mtr*, which calculates MTR using the following formula for each voxel:

$$MTR = \frac{MT0 - MT1}{MT0} * 100 \quad (1)$$

Individual T2-weighted images were then registered to the PAM50 template (*sct_register_to_template* [26]), which was then warped to the native T2-weighted space to define spinal cord regions and tracts with the PAM50 template. MT1 was also registered to a T2-weighted image using *sct_register_multimodal*. This makes group-level and region/tract-based analysis possible. Detailed steps on how MTR images were processed and extracted are described in Supplementary Fig. 1.

Region and tract MTR extraction

For region- and tract-specific MTR extraction, we utilized *sct_extract_metric*, which uses the maximum a posteriori (MAP) method to estimate the maximum likelihood of the regions/tracts [27] and extract more accurate region/tracts-based MTR measures from the PAM50 white matter atlas [26]. The MAP method also helps to mitigate the partial volume effect caused by lower spatial resolution in MT images [23]. MTR measures from the total white matter; ventral, dorsal, and lateral white matter regions; and the ventral corticospinal, ventral reticulospinal, spino-olivary, rubrospinal, lateral corticospinal, lateral reticulospinal, spinal lemniscus, fasciculus gracilis, and fasciculus cuneatus tracts (Supplementary Fig. 2) were extracted from the two vertebral levels containing maximum compression (Supplementary Fig. 1), which was identified on a T2w image. Reported MTR values were averaged across the two vertebral levels and over the regional/tract area.

Statistical analyses

Statistical analysis was performed with GraphPad PRISM version 10.2.2 software. ANOVAs were applied to compare participant characteristics (i.e. age, overall mJOA, and mJOA sub-scores), and region and tract-based MTRs of the HC, mild DCM, and moderate/severe DCM groups. Sample size was based on the results of a previous study, Paliwal et al. [20], which used samples sizes of 9 HCs and 13 DCM patients and observed significant differences between groups. In this study, sample size was increased to 20 HCs and 27 DCM patients to ensure adequate statistical power. Shapiro-Wilk normality tests were applied to the groups before statistical tests were conducted. For data that was normally distributed, a one-way ANOVA with Bonferroni multiple comparison correction was applied. A Kruskal-Wallis test with a Dunns multiple comparison test was applied to data that were not normally distributed. Center values for normally distributed data are represented by the mean and standard deviation (SD), and for non-parametric tests are represented by the median and interquartile range (IQR). Post-hoc corrections were performed separately for each characteristic, region-based, and tract-based comparison. Finally, to evaluate the relationships between the MTR measures and clinical assessment (i.e., mJOA) of DCM

patients, a two-tailed Spearman's correlation was applied due to the non-normal distribution of the data. Differences and correlations with a p-value of <0.05 were considered statistically significant.

RESULTS

Participant characteristics

This cross-sectional study included 27 DCM patients and an age-matched HC group of 20 participants. The mJOA scores among the DCM patients ranged from 8 to 17. DCM patients were categorized into two severity groups based on their mJOA scores: 11 with mJOA scores from 15–17 were categorized into a mild DCM group and 16 with mJOA below 15 were categorized into a moderate and severe (moderate/severe) DCM group. The median ($\pm$ IQR) mJOA score was 16 (± 2.0) for the mild DCM group and 11 (± 3.3) for the moderate/severe DCM group. The mJOA scores were also divided into sub-scores, which included measures for upper limb motor, lower limb motor, upper limb sensory, and sphincter dysfunction. Mild and moderate/severe groups had mean upper limb mJOA scores of 5.0 (± 0.0) and 4.0 (± 1.0), lower limb mJOA scores of 6.0 (± 2.0) and 4.0 (± 0.8), sensory mJOA scores of 2.0 (± 1.0) and 1.0 (± 1.0), and sphincter mJOA scores of 3.0 (± 0.0) and 3.0 (± 1.0), respectively. Statistical testing revealed existing differences in overall and sub-scale mJOA scores between groups (Table 1). The mean ($\pm$ SD) ages for the HC (50.9 ± 6.7), mild DCM (55.91 ± 8.3), and moderate/severe DCM (52.63 ± 5.9) were not statistically significantly different between the groups ($p = 0.16$). Detailed clinical and demographic characteristics of study participants are shown in Table 1.

Comparison of region-based MTR in DCM and HC groups

Significant differences in region-based MTR measures were observed between the HC and moderate/severe DCM groups. In the total white matter, the moderate/severe DCM group had significantly lower MTR compared to the HC ($p = 0.0065$) (Fig. 1A). Similarly, the moderate/severe DCM group had significantly lower MTR in the ventral region compared to the HC group ($p = 0.0009$) (Fig. 1B). However, no significant differences were observed in the lateral ($p = 0.085$) (Fig. 1C) or dorsal ($p = 0.0574$) (Fig. 1D) regions between these two groups. Heatmaps illustrating the average region-based MTR measures (i.e., ventral, dorsal, and lateral regions) are shown in Fig. 1E, which also indicate that greater MTR differences occur within the ventral region. In general, the total white matter and region-based MTR decreased with DCM severity, but there were no significant differences observed between neighboring groups (i.e., HC vs mild DCM and mild DCM vs moderate/severe DCM) in either measures.

Comparison of tract-based MTR in DCM and HC groups

Tract-based MTR comparisons showed results consistent with the regional MTR comparisons. Significant differences in MTR were mainly observed between the HC and moderate/severe DCM groups for tracts located in the ventral region, including the ventral corticospinal ($p = 0.0101$) (Fig. 2A) and ventral reticulospinal ($p = 0.0084$) (Fig. 2B). Significant differences were also found in the spinal lemniscus ($p = 0.0079$) (Fig. 2D), which is a tract located within the lateral region but close to the ventral region. In the dorsal

region, the fasciculus cuneatus ($p = 0.0219$) showed significant difference (Fig. 2H) between the HC and moderate/severe DCM groups. Heatmaps illustrating the average tract-based MTR measures are shown in Fig. 2J, which also indicate that the maximal differences occur within the ventral region. Similar to the region-based results, tract-based MTR generally decreased with DCM severity. An exception to this observation was the rubrospinal tract, which contained an increase in MTR between the mild and moderate/severe DCM groups. Only one significant difference was observed between neighboring groups (i.e., either HC vs mild DCM or mild DCM vs moderate/severe DCM), which was between the HC and mild DCM groups in the spinal lemniscus ($p = 0.0381$) (Fig. 2D).

Association between MTR measures and mJOA scores

At the regional level, no significant correlation ($p > 0.05$) was found between region-based MTR and the overall or sub-scale (i.e., upper limb motor, lower limb motor, upper limb sensory, and sphincter) mJOA measures. The correlations between ventral region MTR values and the overall mJOA (Fig. 3A) and the sub-scale mJOA scores (Fig. 3B–E) were included to compare tract-based results from the ventral region (see discussion below).

Ventral reticulospinal tract MTR was found to be significantly associated with overall mJOA scores ($r = 0.401$; $p = 0.038$) (Fig. 3F), as well as the upper ($r = 0.439$; $p = 0.022$) (Fig. 3G) and lower ($r = 0.386$; $p = 0.047$) (Fig. 3H) limb motor sub-scale scores. However, the ventral reticulospinal MTR measures were not significantly correlated to the upper limb sensory ($r = 0.259$; $p = 0.192$) (Fig. 3I) and sphincter ($r = 0.085$; $p = 0.673$) (Fig. 3J) sub-scale scores. No other significant correlations ($p > 0.05$) between MTR measures and overall/sub-scale mJOA scores occurred, and it is important to note that no other tracts showing significant MTR decreases between groups (Fig. 2A, D, H) were found to correlate to overall or sub-scale mJOA scores. These included the ventral corticospinal (upper limb motor: $r = 0.328$; $p = 0.095$, lower limb motor: $r = 0.146$; $p = 0.469$) tract, the spinal lemniscus (upper limb sensory: $r = -0.045$; $p = 0.822$), and the fasciculus cuneatus (upper limb sensory: $r = 0.293$; $p = 0.138$). It is also noted that no significant correlations were observed between tract-based MTR and the upper limb sensory or sphincter sub-scale mJOA measures. Detailed r - and p -values for the Spearman's correlation between regions displaying significant MTR differences and mJOA scores/sub-scores can be found in Table 2.

DISCUSSION

Degenerative cervical myelopathy (DCM) is a common cause of chronic spinal cord injury in patients. While cord compression is an omnipresent finding in these patients, clinical symptoms of DCM are poorly predicted by standard MRI alone. Previous studies have shown that regional- [17] and tract-specific [19, 20] MTR measures in the spinal cords of DCM patients are lower compared to their healthy counterparts, likely reflecting white matter myelin loss [18]. However, these works exclude key participant groups (i.e. severe DCM patients [20] and healthy controls [19]) and solely employ manual MTR extraction methodologies. The current study seeks to evaluate region- and tract-specific DCM white matter injury and how they result in various levels of clinical disability.

In general, MTR measured at multiple resolutions (i.e., overall white matter, regions, and tracts) decreased with DCM severity (Figs. 1 and 2), indicating the global impact compression has on the spinal cord white matter of DCM patients. More importantly, our present study demonstrates that region- and tract-specific MTR changes within the spinal cord can reflect the inhomogeneous nature of the white matter damage occurring in DCM patients, aligning with other studies demonstrating regional MTRs are impacted differently in the spinal cord white matter [17]. In a previous study done by Hopkins et al., white matter volume was found to significantly decrease in some of the ventral white matter tracts of DCM patients [18]. Our results aligned with Hopkins et al.'s findings and showed that MTR significantly decreased in the ventral region between healthy controls and moderate/severe patients (Fig. 1B). Evidence for the inhomogeneous white matter damage in DCM patients is further revealed in significant tract-specific MTR changes (Fig. 2). Most of the significant MTR decreases in DCM patients occurred in the motor tracts (i.e., the ventral corticospinal and ventral reticulospinal tracts) within the ventral cord, which is consistent with region-based MTR changes. However, no significant decrease in MTR was found in the spino-olivary tract, which is a sensory tract also contained within the ventral cord. Within the lateral and dorsal regions, two sensory tracts (i.e., the spinal lemniscus tract in the lateral region and the fasciculus cuneatus in the dorsal region), exhibited significant MTR changes while the rest did not. This mixture of significant and non-significant tract-specific changes explains the reasoning behind the non-significant region-based MTR decreases occurring within these two regions, which trend towards significance between the HC and moderate/severe DCM groups (lateral: $p = 0.085$; dorsal: $p = 0.0574$, Fig. 1). These observations testify to the benefit of tract-based MTR analysis because they can be used to evaluate the inhomogeneity in white matter damage in progressive DCM. The benefits of tract-based MTR analysis is further supported by the correlation analysis between MTR and mJOA scores, where the only significant correlation was identified from tract-specific MTR (Fig. 3F–H) and not the regional.

Because DCM is a heterogeneous disease with various symptom combinations, we expect tract-specific white matter changes to allow for a better quantification of impairment by allowing us to measure white matter damage occurring in specific motor and sensory tracts. Our most profound findings, that MTR changes within the motor tracts of the ventral region, were consistently observable at multiple spatial resolutions (i.e., regional vs tract). It also reflects a common pattern of injury in DCM patients where initial injury occurs in the ventral white and gray matter [28] and impacts the motor tracts within the region [29]. Motor processes like gait, balance, dexterity, and grip strength are commonly impacted by DCM and have been shown to be significantly decreased when compared to HC/asymptomatic controls [30–34]. The NIH Toolbox Motor Battery has also revealed functional motor differences between DCM and HC groups [35], and correlations with regional- and tract-based MTR values [36]. Furthermore, patient reported outcome measures (PROMIS) upper extremity scores, which has patients self-assess their upper extremity functionality, and PROMIS mobility scores, measuring lower extremity dysfunction, are significantly lower in DCM patients than healthy and asymptomatic controls [37]. Previous findings that establish ventral MTR's correlation to overall mJOA scores [17] also align with our finding of correlations between the ventral reticulospinal tract MTR and overall, upper

and lower limb motor mJOA sub-scores. However, our data allow for a tract-specific look into MTR's correlation with different measures of clinical severity that more closely pertain to the motor roles of the ventral reticulospinal tract.

A factor behind this ventral horn damage could potentially be the degree of cervical kyphosis—a condition in which the curve of the spine inverts – within the spinal column of DCM patients. This can cause the spinal cord to press against anterior part of the vertebrae and lead to tension in the cord, increased intramedullary pressure, and reduced blood supply [38, 39]. T1-slope was also found to impact the size of the C5-C7 neural foramina (i.e. nerve root) [40], which is where common DCM symptoms are suspected to originate, and correlated with cervical lordosis measures and mJOA scores [41]. Furthermore, cervical malalignment can further impact the spinal cord when exposed to dynamic movements, like flexion and extension. For example, flexion can cause the cord to further lengthen and compress against the vertebrae and extension can reduce available cord space and is a risk factor for causing high-intensity lesions on T2w images [38, 39]. Because previous studies have provided evidence for the role that cervical malalignment plays in DCM, complementary imaging, like full length x-rays, dynamic MRI and films, and standing radiographs, when investigating its pathogenesis and progression could be an integral future step in understanding the mechanisms behind white matter damage and symptom manifestation.

Additionally, though only non-significant MTR decreases in the lateral and dorsal regions of DCM patients were noted, tract-based MTR analysis revealed significant changes in the sensory tracts of the dorsal (i.e., the fasciculus cuneatus) and lateral (i.e., the spinal lemniscus, consisting of the spinothalamic and spinoreticular tracts [26]) regions (Fig. 2H, D). Damage to ascending sensory tracts aligns with findings that pain interference PROMIS scores, which measure self-reported pain and its impact on an individual's well-being, were significantly different between DCM patients and healthy/asymptomatic controls [37]. However, correlation between sensory tract MTRs and clinical severity measures were not observed (Fig. 3), which might indicate that sensory tract white matter damage varies between individual patients and is not the result of a consistent injury pattern, like that occurring in the motor tracts of the ventral regions [28, 29]. At the same time, it is also reasonable to observe this damage, as we would expect white matter injury to spread to the lateral and dorsal portions of the cord as myelopathy progresses.

Limitations

This study contains several limitations. Firstly, while the sample size was increased from previous studies [17, 20], sample size is still limited and should be expanded further in future studies in order to gain more robust results for a wider range of DCM patients. Secondly, while significant improvements in spinal cord MRI analysis tools, like the Spinal Cord Toolbox, have been made and the spinal cord can now be automatically segmented, manual corrections were still required in the vast majority of DCM patients due to the spinal cord compression. However, use of automated tools, like the deep learning based spinal cord segmentation, makes segmentation less bias than previous studies using hand drawn masks. Spinal cord imaging technology also still heavily limits data quality. MT imaging

specifically has a low spatial resolution, making it susceptible to the partial volume effect, and MT1 and MT0 images are not acquired simultaneously, which could lead to issues with alignment and MTR calculation. However, advances in analysis methodologies, like the automated registration of MT images to T2w for obtaining regional- and tract-based MTR data based on the PAM50 white matter atlases, helps to limit these effects by providing an image with higher spatial resolution to make data extraction more accurate and reproducible than previous studies. An additional limitation of this imaging technique is the static, supine position of the participants during imaging. Due to this limitation, investigation into the relationship between MTR values and cervical alignment and mobility parameters, which could provide insight into the manifestation and progression of DCM, could not be completed. Finally, while MTR is a validated strategy in quantifying white matter DCM damage, so our analysis was constrained to the white matter tracts and did not include gray matter measurements. Since gray matter damage within the spinal cord is expected to occur alongside white matter loss in DCM patients [28], investigation into gray matter damage is warranted in future studies.

CONCLUSION

DCM is a progressive disease that can cause motor and sensory dysfunction that can ultimately result in permanent neurological damage if left untreated. Currently, surgical decompression is the only corrective treatment; however, surgery is typically reserved for moderate and severe cases, as measured by mJOA. In this study, we established that tract-specific MTR can reflect white matter changes in moderate/severe myelopathy patients that would be lost at the regional/tract level and that upper and lower limb motor mJOA scores correlate with MTR in the ventral reticulospinal tract. This provides evidence that tract-based MTR can reflect white matter damage resulting from cervical spinal cord compression.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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DATA AVAILABILITY

Due to the ongoing nature of data analysis, the datasets generated during and/or analyzed during the current study are not publicly available but are available from the corresponding author on reasonable request.

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Region Based MTR

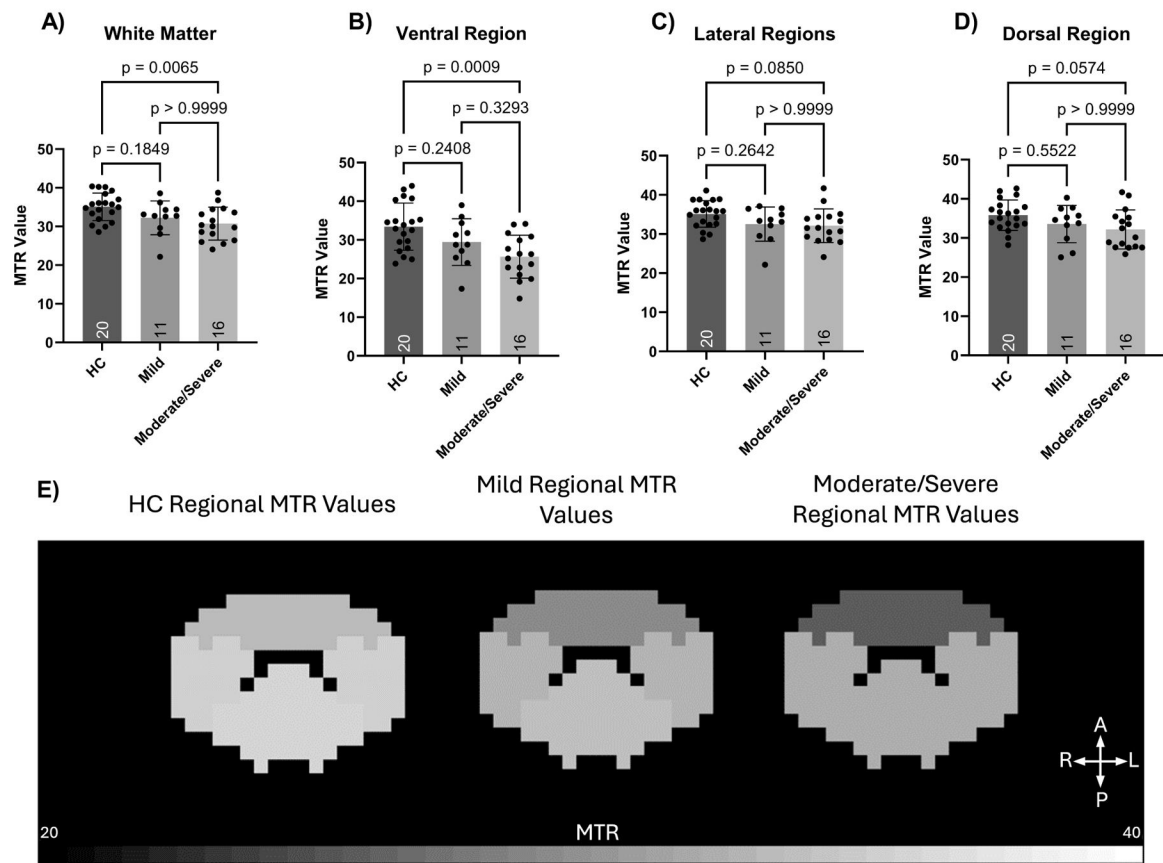


Fig. 1. Region based magnetization transfer ratio (MTR) calculations between healthy controls (HC), and degenerative myelopathy (DCM) patients with mild and moderate/severe mJOA scores.

A one-way ANOVA and a Bonferroni multiple comparison test was used for all comparisons (A–D). Error bars represent the mean and standard deviation of the dataset. Significant differences in MTR values were found between HC and the moderate/severe DCM groups in the white matter ($p = 0.0065$) (A) and ventral ($p = 0.0009$) (B) regions. A heatmap of average regional-specific MTR values (E) depicts differences in HC (left), mild DCM (middle), and moderate/severe DCM (right) participants.

Tract Based MTR

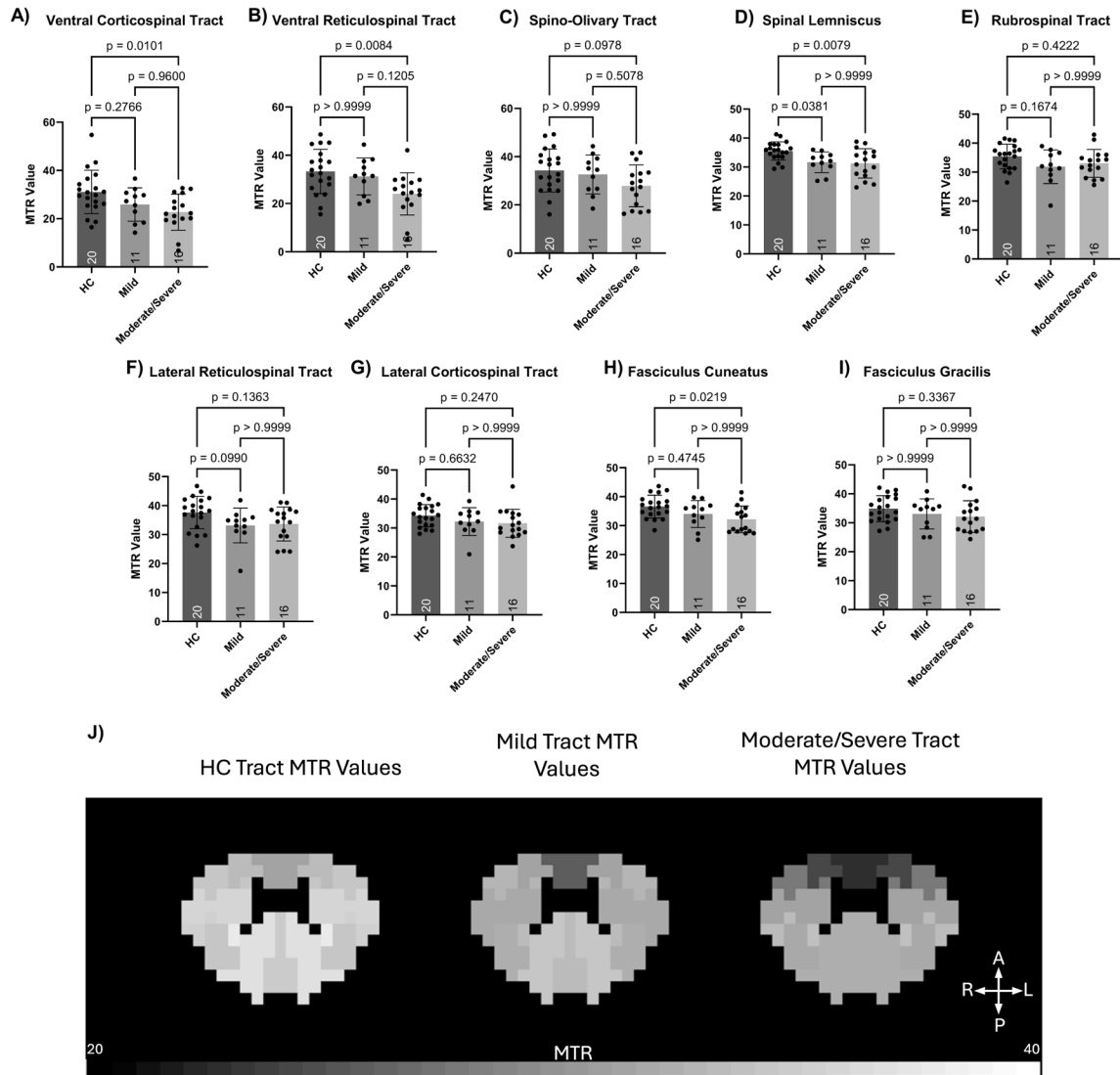
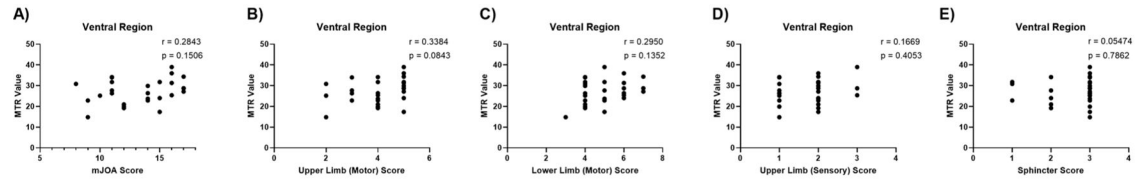


Fig. 2. Tract-based magnetization transfer ratio (MTR) calculations between healthy controls (HC), and degenerative myelopathy (DCM) patients with mild and moderate/severe mJOA scores.

A one-way ANOVA and a Bonferroni multiple comparison test was applied to comparisons for (A–E, G, I), and a Kruskal–Wallis ANOVA and Dunn multiple correction test was applied to comparisons (F) and (H). The error bars represent the mean and standard deviation of the dataset. Significant differences in MTR values were found between HC and the moderate/severe DCM groups in the ventral corticospinal ($p = 0.0101$) and reticulospinal ($p = 0.0084$), and spinal lemniscus ($p = 0.0079$) tracts. Significant differences were also found between the MTR values of HCs and mild DCM groups in the fasciculus cuneatus tract ($p = 0.0219$). A heatmap of average tract-specific MTR values (J) depicts differences in HC (left), mild DCM (middle), and moderate/severe DCM (right) participants.

Correlation Between Ventral Region and mJOA Measures



Correlation Between Ventral Reticulospinal Tract and mJOA Measures

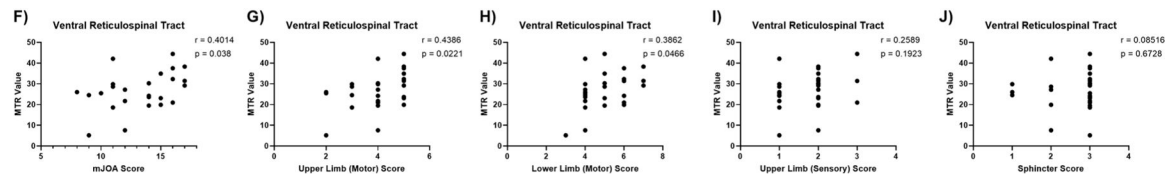


Fig. 3. Correlations between modified Japanese Orthopedic Association (mJOA) measures and magnetization transfer ratio (MTR) values in the ventral region and ventral reticulospinal tract. Correlations between the ventral region (A–E) and ventral reticulospinal tract (F–J) MTR values and mJOA scores and measures. A Spearman’s correlation was used to determine whether any correlations existed between values. There were no significant correlations found, but overall mJOA, upper limb motor, and lower limb motor scores were noted due to tract-specific correlations. Significant correlations were found between the ventral reticulospinal tract MTR values and overall mJOA ($r = 0.4014$; $p = 0.038$) (F), upper limb motor ($r = 0.4386$; $p = 0.0221$) (G), and lower limb motor ($r = 0.3862$; $p = 0.0466$) (H) scores.

Demographic, modified Japanese Orthopedic Association (mJOA), and level of max compression data on the participants grouped in mild, moderate/severe degenerative cervical myelopathy (DCM), and healthy controls (HC).

	Mild DCM (n = 11)	Moderate/Severe DCM (n = 16)	HC (n = 20)	p-value
Age	55.9 ± 8.3	52.6 ± 5.9	50.9 ± 6.7	0.161
Sex (F/M)	8/3	7/9	14/6	N/A
mJOA severity	16 ± 2.0	11 ± 3.3	18 ± 0.0	<0.0001
mJOA sub-scores	Upper limb	5.0 ± 0.0	5.0 ± 0.0	<0.0001
	Lower limb	6.0 ± 2.0	7.0 ± 0.0	<0.0001
	Sensory	2.0 ± 1.0	3.0 ± 0.0	<0.0001
	Sphincter	3.0 ± 0.0	3.0 ± 0.0	0.0018
Level of max compression	C3/C4 (n = 1)	C2/C3 (n = 1)	N/A	N/A
	C4/C5 (n = 5)	C3/C4 (n = 2)		
	C5/C6 (n = 5)	C4/C5 (n = 2)		
		C5/C6 (n = 11)		

Table 2.
Spearman's r-values and their corresponding p-values.

Region/Tract	mJOA score/sub-score	r-value	p-value
White matter	mJOA score	0.1682	0.4017
	Upper limb	0.0889	0.6593
	Lower limb	0.1373	0.4948
	Sensory	0.2256	0.2578
	Sphincter	0.2524	0.2040
Ventral region	mJOA score	0.2843	0.1506
	Upper limb	0.3384	0.0842
	Lower limb	0.2950	0.1352
	Sensory	0.1669	0.4053
	Sphincter	0.0547	0.7862
Ventral reticulospinal	mJOA score	0.4014	0.0380
	Upper limb	0.4386	0.0221
	Lower limb	0.3862	0.0466
	Sensory	0.2589	0.1923
	Sphincter	0.0852	0.6728
Ventral corticospinal	mJOA score	0.1362	0.4983
	Upper limb	0.3281	0.0948
	Lower limb	0.1456	0.4688
	Sensory	0.0176	0.9304
	Sphincter	-0.1095	0.5867
Spinal lemniscus	mJOA score	-0.0687	0.7335
	Upper limb	-0.0766	0.7041
	Lower limb	0.0122	0.9520
	Sensory	-0.0455	0.8219
	Sphincter	0.0852	0.6728
Fasciculus cuneatus	mJOA score	0.2070	0.3002
	Upper limb	0.1319	0.5120
	Lower limb	0.1337	0.5060
	Sensory	0.2928	0.1383
	Sphincter	0.3041	0.1230

Correlation calculations were done between white matter regions displaying significant Magnetization Transfer Ratio (MTR) differences (Figs. 1 and 2) and mJOA scores and sub-scores.

Bolded values represent significant correlations between the region/tract and the mJOA measure.